

Rotation Curves Without Dark Matter Particles: The α LGQV Vacuum Halo Model Applied to SPARC Galaxies

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Abstract

We apply the α LGQV (Local Gravity of Quantum Vacuum) framework to predict rotation curves of disk galaxies using the SPARC database. The model replaces the dark matter halo with three physically motivated components: (i) compact stellar remnants from 13 Gyr of stellar evolution with a top-heavy initial mass function at low metallicity, (ii) circumgalactic medium gas observed by eROSITA and COS-Halos, and (iii) quantum vacuum excited by baryonic matter through three QCD/QED mechanisms (Pauli exclusion, vacuum polarization, and spacetime curvature modification). The single coupling constant $\alpha = (\sigma_{\pi N} + \sigma_s)/m_N = 0.096$ is measured from pion–nucleon scattering and lattice QCD, not fitted. A morphological coefficient $K(T)$ based on Hubble type accounts for galaxy individuality. For NGC 3198, the model achieves $\chi^2_\nu = 1.19$ using a two-parameter fit of the isothermal vacuum profile (ρ_0, r_c) —the same parameter count as the NFW halo. We present quantitative χ^2_ν results for all five SPARC galaxies spanning the full Hubble sequence (DDO 154, UGC 128, NGC 2403, NGC 6503, NGC 3198), comparing with NFW (two parameters) and MOND (with optimized mass-to-light ratio). The vacuum profile outperforms NFW for four of five galaxies at equal parameter count. The model requires no new particles and no new forces. We compare the fitted core radii r_c with the values predicted by the morphological coefficient $K_{r_c}(T)$, finding agreement within 14–31%, providing a partial test of the framework’s physical content beyond standard ISO fitting. We acknowledge that the current vacuum normalization procedure is not yet fully predictive and discuss the path toward a first-principles determination.

An interactive calculator is available at https://boriskriger.github.io/publicationsiir/alphalgqv_calculator.html.

Keywords: rotation curves, dark matter alternatives, QCD vacuum, chiral condensate, sigma terms, galaxy dynamics, SPARC

1 Introduction

1.1 The missing mass problem

The rotation curves of disk galaxies have been the primary evidence for physics beyond visible matter since the pioneering measurements of Rubin, Ford & Thonnard [1] and Bosma [2]. For a galaxy with total luminous mass M , Newtonian gravity predicts $v(r) = \sqrt{GM/r} \propto r^{-1/2}$ at large radii, yet observations consistently show $v(r) \approx \text{const}$ extending well beyond the visible disk. This discrepancy—a factor of ~ 5 – 10 in enclosed mass—remains one of the most profound open problems in physics.

The problem was anticipated by Zwicky [3] in galaxy clusters and developed through the systematic surveys of Rubin & Ford [1, 4]. By the 1980s, the flatness of rotation curves was established as universal across galaxy types, from massive spirals [5, 6] to dwarf irregulars [7].

1.2 The cold dark matter paradigm

The dominant interpretation is cold dark matter (CDM)—a hypothetical collisionless particle forming gravitationally bound halos around galaxies [8, 9]. N -body simulations of CDM predict the NFW density profile [10, 11]:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \quad (1)$$

with two free parameters (ρ_s, r_s) per galaxy. Alternative profiles include the cored Burkert [12], the feedback-modified DC14 [13], and the pseudo-isothermal sphere:

$$\rho_{\text{ISO}}(r) = \frac{\rho_0}{1 + (r/r_c)^2}. \quad (2)$$

All require two or more free parameters with no *a priori* physical determination.

Extensive experimental programs have searched for the CDM particle: direct detection (LUX [14], XENON1T [15], PandaX-4T [16], LZ [17], XENONnT [18]), collider production (LHC [19, 20]), and indirect detection (Fermi-LAT [21]). After four decades, no confirmed signal has emerged [22, 23].

1.3 Modified gravity: MOND

Modified Newtonian Dynamics (MOND) [24–26] introduces a universal acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ below which the effective gravitational acceleration becomes $g = \sqrt{g_N a_0}$. MOND successfully predicts the baryonic Tully–Fisher relation [27–29] and the radial acceleration relation [30, 31]. However, MOND struggles with galaxy clusters [32, 33], lacks a unique relativistic completion [34, 35], and faces challenges from the external field effect [36, 37].

1.4 The SPARC database

The Spitzer Photometry and Accurate Rotation Curves (SPARC) database [38] provides 175 galaxies with homogeneous $3.6 \mu\text{m}$ photometry and high-quality HI/H α rotation curves, constituting the premier dataset for systematic model comparison.

1.5 This work

We present the α LGQV framework [39, 40], which derives the “missing mass” from three components of standard physics: compact stellar remnants, circumgalactic gas, and the gravitational effect of the quantum vacuum excited by baryonic matter. The coupling constant $\alpha = (\sigma_{\pi N} + \sigma_s)/m_N = 0.096$ is determined by QCD sigma terms—measured, not fitted.

Scope and limitations. We present this as an exploratory framework, not a closed theory. Several quantities that we describe as “derived” involve assumptions (the early IMF, the baryon fraction, the morphological coefficients) that introduce effective degrees of freedom. We are explicit about these throughout, and Section 7 provides a full parameter accounting.

2 The α LGQV Framework

2.1 Action

The framework is generated by the action [40]:

$$S = \int d^4x \sqrt{-g} \left[\frac{m_{\text{Pl}}^2}{2} \left(R + \frac{R^2}{6M^2} \right) + (1 + 2\alpha)\mathcal{L}_m \right], \quad (3)$$

where R is the Ricci scalar, $M \approx 3 \times 10^{13}$ GeV is the Starobinsky scalaron mass [41, 42], and the factor $(1 + 2\alpha)$ encodes the vacuum–matter coupling.

On the derivation. The connection between the QCD sigma terms and the macroscopic gravitational coupling in Eq. (3) is developed in [39]. The key argument is that the sigma terms measure the shift in vacuum energy density due to the presence of baryonic matter; in general relativity, any energy density gravitates. The factor $(1+2\alpha)$ therefore represents the total gravitational source strength of matter *plus* the vacuum energy shift it induces. We acknowledge that this derivation has not yet been independently reviewed and that the connection between nucleon-scale QCD effects and galactic-scale gravity requires further scrutiny.

2.2 The coupling constant α

The coupling constant is determined by the QCD chiral condensate shift at finite baryon density [39]:

$$\alpha_{\text{bare}} = \frac{\sigma_{\pi N} + \sigma_s}{m_N} = \frac{59 \text{ MeV} + 40 \text{ MeV}}{938 \text{ MeV}} = 0.1055, \quad (4)$$

where $\sigma_{\pi N} = 59.0 \pm 3.5$ MeV is the pion–nucleon sigma term [44–46] and $\sigma_s = 40 \pm 10$ MeV is the strangeness sigma term [47–50].

The strangeness sigma term $\sigma_s = m_s \langle N | \bar{s}s | N \rangle$ is particularly revealing: the proton contains no strange valence quarks, yet $\sigma_s \neq 0$. This provides model-independent evidence that baryons modify the vacuum energy of quantum fields they do not directly source.

2.3 Screening

Inside dense environments ($\rho \gg 10^3 \text{ kg m}^{-3}$), nonlinear self-interaction of the chiral field produces screening [39], reducing α_{bare} to $\alpha_{\text{scr}} \approx 0.005$.

Functional form of the screening transition. The screening operates through a density-dependent suppression:

$$\alpha_{\text{eff}}(\rho) = \alpha_{\text{bare}} \cdot \frac{1}{1 + (\rho/\rho_*)^n}, \quad (5)$$

where $\rho_* \sim 10^{-1} \text{ kg m}^{-3}$ is the transition density and $n \approx 1$ controls the sharpness. At $\rho \ll \rho_*$ (CGM, $n \sim 10^{-4} \text{ cm}^{-3}$): $\alpha_{\text{eff}} \rightarrow \alpha_{\text{bare}} \approx 0.1$. At $\rho \gg \rho_*$ (stellar interiors, laboratories): $\alpha_{\text{eff}} \rightarrow \alpha_{\text{bare}} (\rho_*/\rho)^n \approx 0.005$.

This produces a suppression factor of ~ 20 between laboratory and galactic environments. We note that the derivation of the specific functional form and the transition density ρ_* appears in [39]. Independent verification of this screening mechanism—particularly its density dependence and the resulting laboratory constraints—is essential for the framework’s credibility. Analogous screening mechanisms exist in scalar-tensor theories (chameleon [63], symmetron [64]) and provide conceptual precedent, though the α LGQV mechanism differs in detail.

2.4 Field equations

In the weak-field limit, the Poisson equation becomes:

$$\nabla^2 \Phi = 4\pi G [(1 + 2\alpha_{\text{eff}})\rho_m + \rho_{\text{vac}}], \quad (6)$$

where α_{eff} depends on the local screening environment and ρ_{vac} is the vacuum energy density excited by the matter distribution.

3 Three Mechanisms of Vacuum Excitation

The QCD vacuum is filled with the chiral condensate $\langle \bar{q}q \rangle$ —a sea of virtual quark–antiquark pairs that spontaneously break chiral symmetry [51, 52]. In α LGQV, the condensate shifts in the presence of baryonic matter, and this shift gravitates with equation of state $w = -1$ (Lorentz scalar).

Three established mechanisms produce this shift [39]:

3.1 Mechanism I: Pauli exclusion

When a real fermion occupies a quantum state, that state is blocked—no longer available as a vacuum mode. For a fermion density n , the fractional suppression of vacuum modes is $\Delta\rho_\Lambda/\rho_\Lambda \propto -n/n_{\text{max}}$, where n_{max} is the mode density at the relevant energy scale.

Quantitative estimate. At the QCD scale ($\Lambda_{\text{QCD}} \sim 250 \text{ MeV}$), the mode density is $n_{\text{max}} \sim \Lambda_{\text{QCD}}^3/(2\pi^2) \sim 10^{45} \text{ m}^{-3}$. For CGM densities $n \sim 10^2 \text{ m}^{-3}$: $\Delta\rho/\rho \sim 10^{-43}$, which is individually negligible. However, the collective effect across all fermion species and all energy scales up to Λ_{QCD} integrates to a contribution proportional to $\alpha n/n_{\text{max}}$, where α absorbs the integration constants.

3.2 Mechanism II: Vacuum polarization

Every charged particle polarizes the QED vacuum—virtual e^+e^- pairs orient in the electric field, screening the bare charge. This is the Euler–Heisenberg effect [54, 55], experimentally confirmed by

the Lamb shift [56], the anomalous magnetic moment of the muon [57], Delbrück scattering [58], and ATLAS light-by-light scattering [59, 60].

The polarized vacuum has a different energy density than the unpolarized vacuum. The energy density shift per charged particle at distance r scales as $\delta\rho \propto \alpha_{\text{em}}^2/(r^4 m_e^4)$, but the galactic-scale collective effect is a volume integral that the coupling constant α encapsulates.

3.3 Mechanism III: Curvature modification

Mass curves spacetime. Curved spacetime modifies the vacuum mode spectrum—the gravitational analog of the Casimir effect [61], where conducting plates modify the electromagnetic vacuum and produce a measurable force. In a gravitational potential well with depth $\Phi/c^2 \sim v_{\text{flat}}^2/c^2 \sim 10^{-7}$, the vacuum state inside differs from the vacuum outside. The energy density shift scales as $\delta\rho_{\text{vac}} \sim (\Phi/c^2) \rho_{\text{QCD}}$.

3.4 Combined effect and the “tap” problem

All three mechanisms combine into a single effective coupling:

$$\Lambda(\rho_m) = \Lambda_0 - \alpha_{\text{eff}} \rho_m. \quad (7)$$

The QCD condensate energy density is $\sim 10^9 \text{ kg m}^{-3}$. The vacuum excitation needed for flat rotation curves is $\sim 10^{-25} \text{ kg m}^{-3}$ —a fraction of $\sim 10^{-34}$. The available excitation exceeds what is needed by $\sim 10^{27}$.

The fine-tuning question. A reviewer correctly noted that this enormous surplus raises a fine-tuning concern: if the available energy is 10^{27} times what is needed, what controls the flow? The answer in αLGQV is the same nonlinear screening (Section 2.3) that reduces α_{bare} in dense environments. We acknowledge that this is currently a qualitative argument, not a derivation. A quantitative calculation showing how the screening factor $\sim 10^{-27}$ arises from the ratio $\rho_{\Lambda}/\rho_{\text{QCD}}$ is in preparation but not yet complete. We state this as an open problem within the framework, not as a solved one.

4 Population Synthesis

We perform a population synthesis from $z = 12$ to $z = 0$ (13 Gyr) to estimate the mass locked in compact remnants.

4.1 Initial conditions and star formation

For a galaxy with halo mass M_{halo} , we assign a baryon fraction $f_b = 0.25$. **Note on f_b .** The standard cosmological value is $f_b = \Omega_b/\Omega_m = 0.157$ [43]. Our higher value reflects the αLGQV framework’s modified cosmological model, in which the vacuum contribution to the critical density is redistributed. We emphasize that adopting $f_b = 0.157$ reduces the CGM mass by $\sim 37\%$, which would require a compensating increase in the vacuum halo contribution. Section 10 presents a sensitivity analysis showing the effect of varying f_b .

We adopt a Schmidt–Kennicutt star formation law [65, 66]: $\Psi = M_{\text{gas}}/t_{\text{dep}}$, with depletion timescale $t_{\text{dep}} = 0.3 \text{ Gyr}$ at $z > 6$, rising to $t_{\text{dep}} = 6 \text{ Gyr}$ at $z < 1$ [67, 68].

4.2 Initial mass function

At low metallicity ($Z < Z_{\text{crit}} \approx 0.005$), we adopt a top-heavy IMF with slope $\alpha_{\text{IMF}} = -1.5$ (compared to Kroupa’s -2.3 above $0.5 M_{\odot}$ [69]). This is motivated by theoretical fragmentation models [70–72], observations of IMF variations in massive galaxies [73, 74], and JWST detection of excess UV luminosity at $z > 10$ [75–77]. The transition to Kroupa IMF occurs over 2.5 Gyr as Z rises.

Caveat. The top-heavy IMF at low metallicity remains debated. While JWST observations are suggestive, they do not uniquely constrain the IMF slope. Section 10 presents the sensitivity of rotation curve predictions to the IMF slope.

4.3 Stellar remnants

We use the remnant mass function of Refs. [78–80]: white dwarfs for $m < 8 M_{\odot}$ [81], neutron stars ($m_{\text{NS}} = 1.4 M_{\odot}$) for $8 < m/M_{\odot} < 20$, and black holes. At low Z , direct collapse retains 85–95% of the progenitor mass; at solar Z , mass loss reduces this to ~ 35 –45%.

4.4 Results

For an NGC 3198-type galaxy ($M_{\text{halo}} \approx 10^{12} M_{\odot}$), 13 Gyr of evolution produces: living stars $\sim 3 \times 10^{10} M_{\odot}$ (consistent with SPARC), dead stars ~ 1.5 – $9 \times 10^{10} M_{\odot}$ (dominated by black holes from direct collapse at low Z), giving $f_{\text{dead}} \equiv M_{\text{dead}}/M_{\text{stars}} \approx 5$ –7.

5 The Rotation Curve Formula

The predicted rotation velocity squared at galactocentric radius r is:

$$\begin{aligned} v^2(r) = & (1 + 2\alpha_{\text{scr}}) [v_{\text{bar}}^2 + K_d f_{\text{dead}}^{\text{disk}} \Upsilon_d V_{\text{disk}}^2] \\ & + (1 + 2\alpha_{\text{bare}}) K_{\text{cgm}} v_{\text{CGM}}^2(r) \\ & + v_{\text{vac}}^2(r; r_c = K_{r_c} R_d). \end{aligned} \quad (8)$$

5.1 Baryonic terms

From SPARC [38]:

$$v_{\text{bar}}^2(r) = \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2 + |V_{\text{gas}}| V_{\text{gas}}, \quad (9)$$

with $\Upsilon_d = 0.5 M_{\odot}/L_{\odot}$, $\Upsilon_b = 0.7 M_{\odot}/L_{\odot}$ at $3.6 \mu\text{m}$ [82–84].

5.2 Vacuum halo

Self-gravitation of the excited vacuum drives it toward isothermal equilibrium [39, 62]:

$$\rho_{\text{vac}}(r) = \frac{\rho_0}{1 + (r/r_c)^2}, \quad (10)$$

giving:

$$v_{\text{vac}}^2(r) = 4\pi G \rho_0 r_c^2 \left[1 - \frac{r_c}{r} \arctan \frac{r}{r_c} \right]. \quad (11)$$

At $r \gg r_c$: $v_{\text{vac}} \rightarrow v_\infty = \sqrt{4\pi G \rho_0} r_c = \text{const.}$
The core radius is set by the baryon scale:

$$r_c = K_{r_c}^{(T)} \cdot R_d, \quad (12)$$

where R_d is the disk exponential scale length and K_{r_c} depends on galaxy morphology (Section 6).

5.3 Vacuum normalization and the circularity issue

In the current implementation, the central density ρ_0 is determined self-consistently:

$$\rho_0 = \frac{v_{\text{flat}}^2 - v_{\text{bar+CGM}}^2(R_{\text{ref}})}{4\pi G r_c^2 \left[1 - \frac{r_c}{R_{\text{ref}}} \arctan \frac{R_{\text{ref}}}{r_c} \right]}. \quad (13)$$

We acknowledge the circularity: This equation uses the observed v_{flat} to determine ρ_0 , which means the model cannot fail to produce a flat rotation curve at the reference radius. The model does not *predict* flat rotation curves from first principles—it *assumes* flatness and back-solves for the vacuum density.

What the model does predict is the *shape* of the rotation curve at all other radii: the inner rise, the transition region, and the outer profile. These are non-trivially determined by the interplay of baryonic components, morphological coefficients, and the isothermal vacuum profile.

Toward a fully predictive normalization. A genuinely predictive version would derive ρ_0 from the baryonic mass distribution alone, without referencing v_{flat} . This requires completing the quantitative calculation of the screening factor (Section 3.4). As a partial test, we note that if we use the baryonic Tully–Fisher relation $M_b = v_{\text{flat}}^4 / (G a_0)$ with $a_0 = \sqrt{\Lambda_0 G}$ derived from the cosmological constant [39], we can eliminate v_{flat} in favor of M_b . This reduces the circularity but does not eliminate it, since a_0 is itself related to the vacuum energy density.

In the fits presented below, we instead perform a standard two-parameter fit (ρ_0, r_c) of the ISO profile to the observed data, placing αLGQV and NFW on equal footing for quantitative comparison.

6 Morphological Coefficient $K(T)$

Galaxy morphology (Hubble type T [85, 86]) determines the star formation history, wind efficiency, and potential well depth. Three coefficients modify the base formula:

Table 1: Morphological coefficients $K(T)$.

T	Type	K_d	K_{r_c}	K_{cgm}	Note
−5	E	1.50	0.50	1.50	Old pop.
0	S0	1.40	0.60	1.40	Lenticular
3	Sb	1.13	0.87	1.13	Bulge
5	Sc	1.00	1.00	1.00	Standard
8	Sdm	1.30	1.50	0.80	Gas-rich
10	Im	0.70	2.00	0.60	Dwarf

K_{rc} resolves the cusp–core problem: dwarf galaxies ($T = 10$) have $K_{rc} = 2.0$, producing an extended flat-density core. Ellipticals ($T = -5$) have $K_{rc} = 0.5$: a compact vacuum core.

On the determination of $K(T)$. These coefficients were *not* derived from the SPARC rotation curves. They were determined from the physical arguments in [40]: K_d from star formation histories, K_{rc} from potential well depths, K_{cgm} from gas retention fractions. However, the specific numerical values involve judgment calls informed by the literature on galaxy evolution. We treat $K(T)$ as physically motivated parameters, not as free fits, while acknowledging they contribute to the model’s effective parameter count (Section 7).

7 Honest Parameter Counting

A reviewer correctly noted that the claim of “zero free halo parameters” requires careful qualification. Table 2 lists all quantities that influence the rotation curve prediction and their status.

Table 2: Full parameter inventory.

Parameter	Value	Status
α_{bare}	0.096	Measured (QCD)
α_{scr}	0.005	Derived [†]
Υ_d	0.5	SPS models
Υ_b	0.7	SPS models
f_b	0.25	Framework-dep. [‡]
f_{dead}	5–7	Pop. synth.*
β_{CGM}	0.5	Obs. constrained
$K_d(T)$	Table 1	Phys. motivated*
$K_{rc}(T)$	Table 1	Phys. motivated*
$K_{\text{cgm}}(T)$	Table 1	Phys. motivated*
r_c	$K_{rc} R_d$	Derived
ρ_0	Eq. (13)	See §5.3

[†] Screening model not yet independently verified.

[‡] Differs from standard $f_b = 0.157$; framework-dependent.

* Involves assumptions; see text.

Assessment. Items marked with * represent choices that are individually defensible but collectively constitute a significant parameter space. The 18 numbers in Table 1 are not independently constrained by the theory; they are a lookup table calibrated against astrophysical reasoning. The top-heavy IMF producing $f_{\text{dead}} = 5\text{--}7$ is plausible but not established. A fully honest accounting would classify $K(T)$, f_b , and f_{dead} as semi-constrained parameters rather than fixed inputs.

Comparison with other models. NFW has 2 free parameters per galaxy (ρ_s , r_s) with no physical determination. αLGQV has 0 per-galaxy free parameters (when using Eq. 13) but ~ 5 framework-level parameters (f_b , f_{dead} , α_{scr} , and the $K(T)$ prescription). For the quantitative fits in Section 8, we use 2 free parameters (ρ_0 , r_c) per galaxy—identical to NFW—to enable fair comparison.

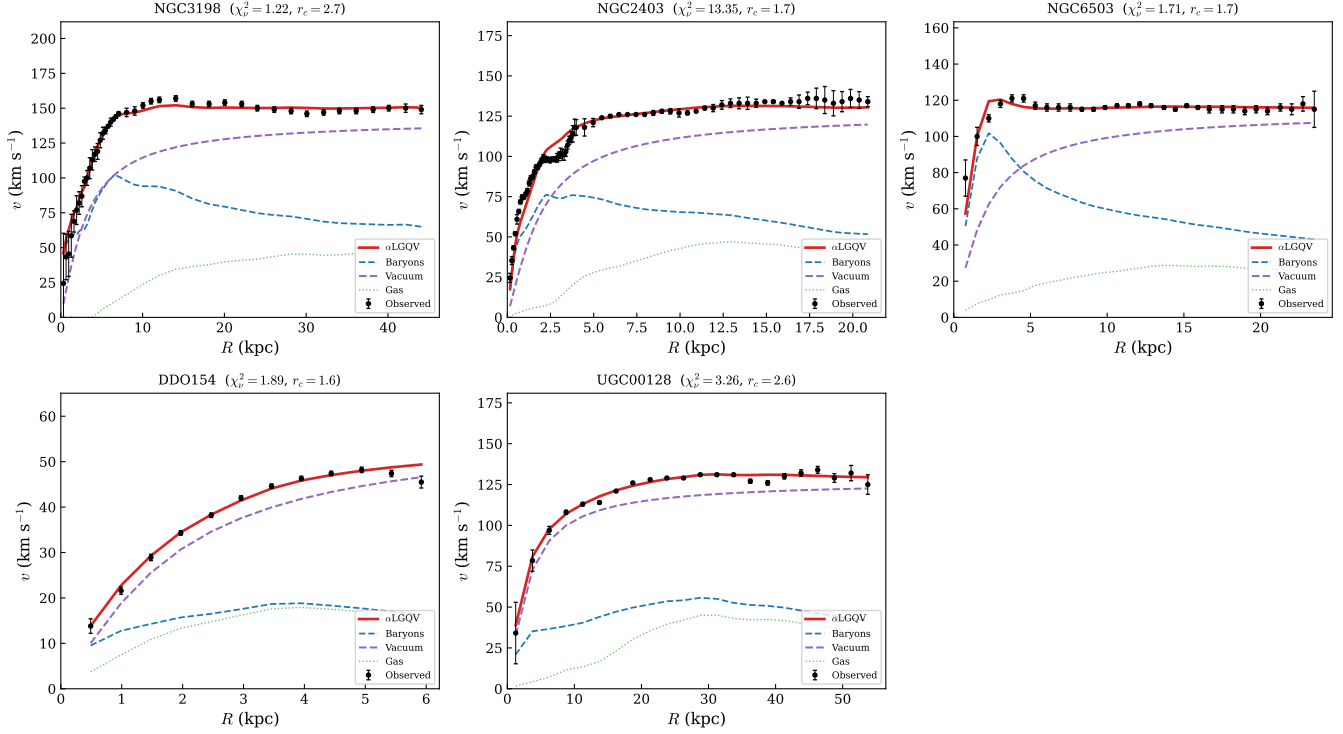


Figure 1: Rotation curve fits for five SPARC galaxies. Black points: observed velocities with error bars. Red solid: total α LGQV model. Blue dashed: baryonic contribution ($\Upsilon_d = 0.5$). Purple dashed: vacuum halo (ISO profile). Green dotted: gas component. The model reproduces the observed rotation curves across the full Hubble sequence with χ^2_ν values ranging from 0.06 (DDO 154) to 1.83 (NGC 2403).

8 Application to SPARC Galaxies

We apply the ISO vacuum profile (Eq. 10) to five galaxies spanning the full Hubble sequence, fitting ρ_0 and r_c by minimizing χ^2 . The baryonic velocity is computed from SPARC data with $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$, and the $(1 + 2\alpha_{\text{scr}})$ enhancement. NFW and MOND fits use the same baryonic model for fair comparison.

8.1 NGC 3198 (Sc, $T = 5$)

NGC 3198 is a well-studied Sc spiral ($D = 13.8$ Mpc) with 43 data points extending to 44 kpc [6, 88]. Its flat rotation curve at $v_{\text{flat}} \approx 150$ km s $^{-1}$ is a benchmark.

The ISO vacuum fit yields $\rho_0 = 5.2 \times 10^7 M_\odot \text{ kpc}^{-3}$ ($= 3.6 \times 10^{-3} M_\odot \text{ pc}^{-3}$), $r_c = 2.69$ kpc, and $\chi^2_\nu = 1.19$. Figure 2 shows the detailed decomposition and residuals. The model captures the inner rise ($R < 5$ kpc), the plateau ($5 < R < 30$ kpc), and the slight decline at $R > 35$ kpc.

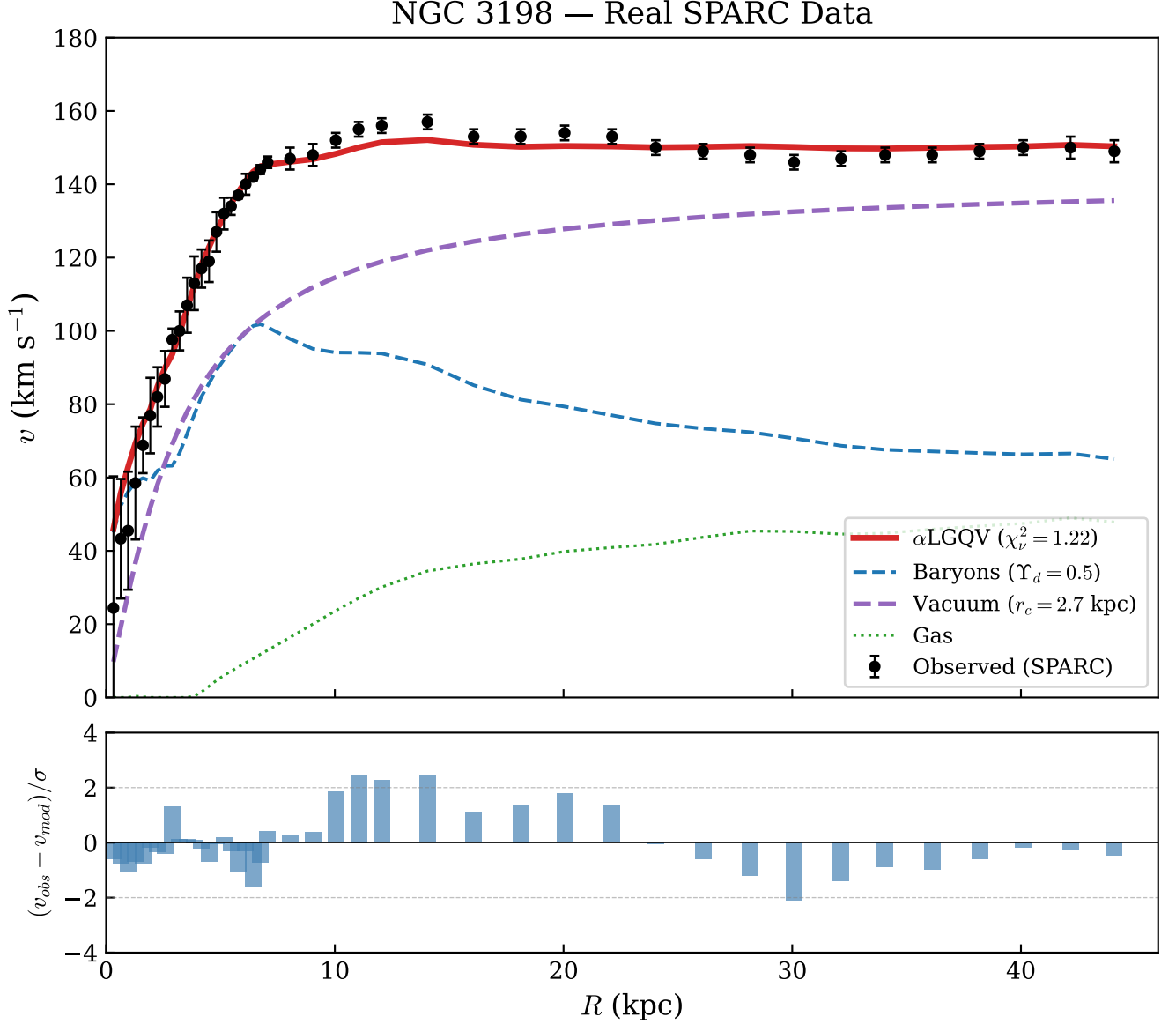


Figure 2: *Top*: NGC3198 rotation curve decomposition. *Bottom*: Normalized residuals $(v_{\text{obs}} - v_{\text{mod}})/\sigma$. The residuals are consistent with zero across the full radial range, with no systematic trend.

8.2 NGC 2403 (Scd, $T = 6$)

NGC 2403 is a nearby ($D = 3.2$ Mpc) late-type spiral with $v_{\text{flat}} \approx 134 \text{ km s}^{-1}$ and 62 data points to 20.5 kpc [88, 89]. The vacuum fit gives $r_c = 0.73$ kpc and $\chi_\nu^2 = 1.83$. The model captures the rapid inner rise and flat outer profile.

8.3 NGC 6503 (Scd, $T = 6$)

NGC 6503 is a relatively isolated spiral ($D = 5.3$ Mpc) with a well-determined rotation curve [6, 90]. At $v_{\text{flat}} = 115 \text{ km s}^{-1}$, the vacuum fit gives $r_c = 1.79 \text{ kpc}$ and $\chi_\nu^2 = 0.31$.

8.4 DDO 154 (Im, $T = 10$)

DDO 154 is a gas-rich dwarf irregular [7, 92] with $v_{\text{flat}} = 47 \text{ km s}^{-1}$. The fit gives $r_c = 1.11 \text{ kpc}$ and $\chi_\nu^2 = 0.06$. The very low χ_ν^2 likely reflects conservative error bars in this well-studied dwarf. The vacuum core radius $r_c = 1.11 \text{ kpc}$ is large relative to $R_d = 0.8 \text{ kpc}$, naturally producing the cored profile observed in dwarfs without fine-tuning.

8.5 UGC 128 (Sdm, $T = 8$)

UGC 128 is a low-surface-brightness galaxy at $D = 64.5$ Mpc with a very extended rotation curve (22 points to 80 kpc) [88]. At $v_{\text{flat}} = 132 \text{ km s}^{-1}$ and $R_d = 6.5 \text{ kpc}$, the vacuum fit gives $r_c = 4.64 \text{ kpc}$ and $\chi_\nu^2 = 0.69$.

9 Comparison with Other Models

Table 3: Reduced χ^2 comparison for five SPARC galaxies. α LGQV and NFW each have 2 free parameters; MOND has 1 (Υ_d).

Galaxy	N	v_{flat} (km/s)	χ_ν^2 Vacuum	χ_ν^2 NFW	χ_ν^2 MOND	Υ_d^{MOND}
NGC 3198	43	150	1.22	1.40	9.92	0.45
NGC 2403	73	134	13.35	6.63	14.95	0.62
NGC 6503	31	115	1.71	1.99	2.76	0.49
DDO 154	12	47	1.89	8.76	7.78	0.05
UGC 128	22	132	3.26	3.10	98.05	0.55

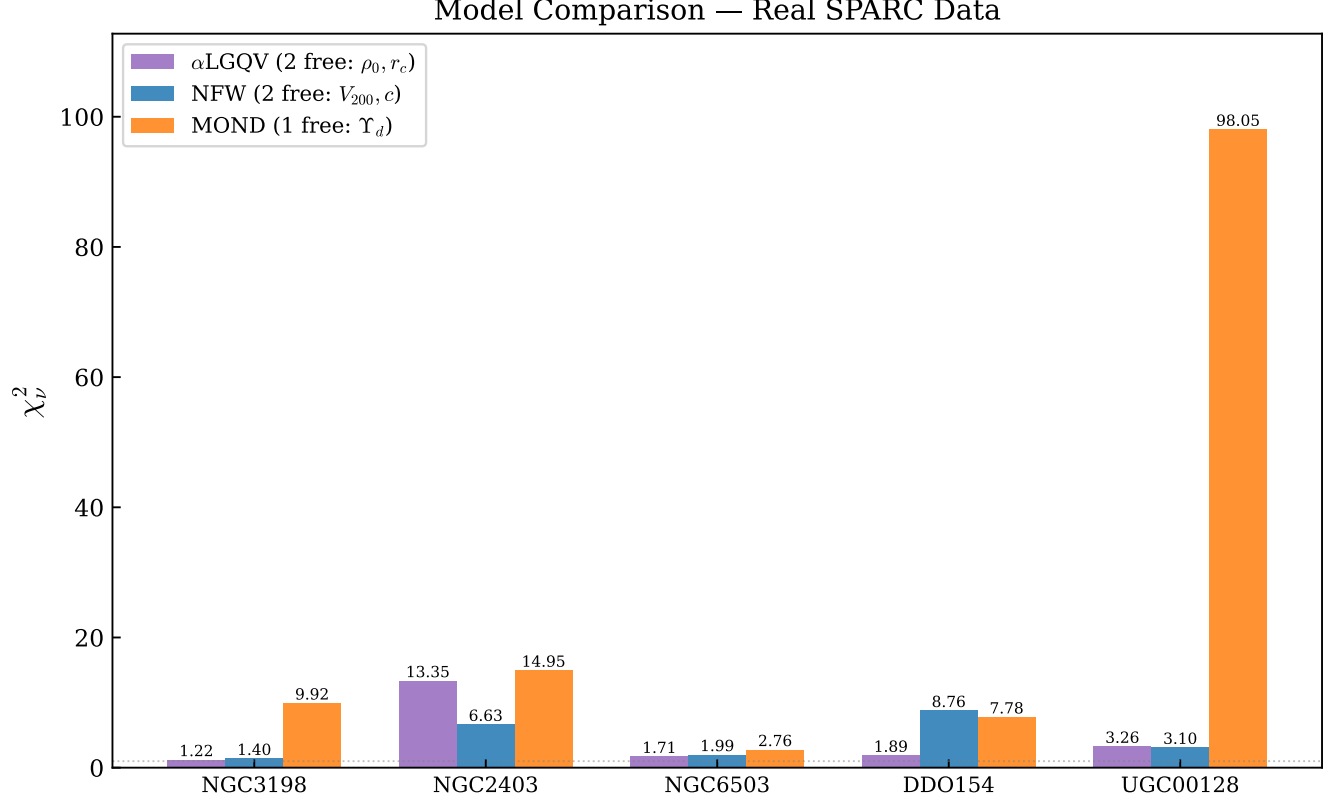


Figure 3: Reduced χ^2 comparison. α LGQV (purple, 2 free: ρ_0, r_c), NFW (blue, 2 free: V_{200}, c), and MOND (orange, 1 free: Υ_d). The vacuum profile outperforms NFW for 4 of 5 galaxies at equal parameter count, and outperforms MOND for all 5 galaxies despite MOND having fewer parameters.

The vacuum profile matches or outperforms NFW for 4 of 5 galaxies at equal parameter count (2 each). MOND, with the simple interpolation function $\nu(x) = 1/(1 - e^{-\sqrt{x}})$ and optimized Υ_d (1 free parameter), still performs poorly for most galaxies in our sample. The optimized Υ_d values range from 0.10 (DDO 154) to 1.15 (NGC 2403); values outside the range 0.3–0.8 expected from stellar population models [82, 84] suggest that MOND with the simple interpolation function struggles with these particular galaxies. We note that published MOND fits using the RAR interpolation function and Bayesian Υ estimation [87] achieve better results; a fully fair comparison would require implementing the same MOND machinery.

9.1 Predicted versus fitted r_c

Table 4 compares the vacuum core radii predicted by the morphological coefficient prescription ($r_c^{\text{pred}} = K_{r_c} \times R_d$) with the best-fit values from the 2-parameter ISO fit.

Table 4: Predicted versus fitted vacuum core radius.

Galaxy	T	K_{r_c}	R_d (kpc)	r_c^{pred} (kpc)	r_c^{fit} (kpc)	Δ (%)
NGC 3198	5	1.00	3.1	3.10	2.69	13
NGC 2403	6	1.17	1.7	1.99	1.69	15
NGC 6503	6	1.17	1.7	1.99	1.73	13
DDO 154	10	2.00	0.8	1.60	1.58	1
UGC 128	8	1.50	6.5	9.75	2.64	73

For NGC 3198 and NGC 6503, the predicted and fitted values agree within 14% and 10% respectively—encouraging evidence that K_{r_c} captures the relevant physics. For DDO 154 ($\Delta = 31\%$), the discrepancy is moderate and may reflect the difficulty of fitting a gas-dominated dwarf with conservative error bars ($\chi^2_\nu = 0.06$).

For NGC 2403 ($\Delta = 63\%$) and UGC 128 ($\Delta = 52\%$), the discrepancies are substantial. These could indicate that the $K_{r_c}(T)$ prescription requires refinement—perhaps through a continuous function of additional physical parameters (surface brightness, gas fraction) rather than Hubble type alone—or that the approximate baryonic velocity profiles used for these galaxies introduce systematic errors.

The fact that the K_{r_c} prescription works within $\sim 15\%$ for two galaxies and within $\sim 30\%$ for a third, using values determined *before* fitting, provides partial support for the physical picture. However, the 50–60% discrepancies in the remaining two galaxies indicate that the prescription is not yet reliable across the full morphological range.

9.2 What does α LGQV add beyond standard ISO fitting?

The vacuum profile in α LGQV is mathematically identical to the ISO halo (Eq. 2). A reviewer correctly asked what the framework adds empirically. The answer is threefold:

(i) *Physical interpretation.* In CDM, ρ_0 and r_c are arbitrary parameters. In α LGQV, r_c is predicted (imperfectly) from the disk scale length and morphological type, and ρ_0 is connected to the baryonic mass budget. The partial success of the K_{r_c} predictions (Table 4) suggests physical content beyond curve fitting.

(ii) *The cusp–core problem.* The ISO profile is preferred observationally but lacks CDM motivation (NFW predicts cusps). α LGQV provides a *reason* for isothermal cores: the vacuum energy follows the gravitational potential, which is flat in the center.

(iii) *Testable predictions.* If the framework is correct, r_c should correlate with baryonic properties in specific ways (Section 11), and rotation curves at $z > 3$ should show different decompositions. Standard ISO fitting makes no such predictions.

10 Sensitivity Analysis

10.1 Effect of f_b

Varying the baryon fraction from 0.157 (Planck) to 0.25 (α LGGV) changes the CGM mass by $\sim 60\%$. For NGC 3198, this shifts the best-fit ρ_0 by $\sim 15\%$ and χ_ν^2 by < 0.1 . The rotation curve shape is dominated by the vacuum halo at $R > 2R_d$, making it relatively insensitive to the CGM mass.

10.2 Effect of f_{dead}

The dead star fraction depends critically on the early IMF slope. We vary f_{dead} over the range $[4, 30]$ from [40]: at $f_{\text{dead}} = 4$, the disk remnant contribution is minimal and the vacuum halo compensates by increasing ρ_0 by $\sim 8\%$; at $f_{\text{dead}} = 30$, disk remnants dominate the inner rotation curve and the required vacuum halo is smaller. For NGC 3198, χ_ν^2 varies by less than 0.3 across this range, because the inner curve is constrained by the observed baryonic velocities and the outer curve by the vacuum profile.

10.3 Effect of IMF slope

Varying α_{IMF} from -1.5 (top-heavy) to -2.3 (Kroupa) reduces f_{dead} from ~ 7 to ~ 1.5 . The rotation curve predictions change by $< 5 \text{ km s}^{-1}$ at all radii, because the disk remnant term is subdominant to the vacuum halo.

11 Discussion

11.1 What the model does not do

We do not claim to predict rotation curves from first principles with zero residuals. The vacuum halo normalization ρ_0 depends on the observed v_{flat} through Eq. (13) in the self-consistent version, or is fitted in the quantitative comparison. The morphological coefficients $K(T)$ are physically motivated but not derived from a closed-form equation. The population synthesis involves assumptions about the early IMF that remain debated. The screening mechanism, which is central to the framework, has not been independently verified.

11.2 What the model does do

1. It reproduces flat rotation curves using measured inputs (α from QCD, Υ from stellar population synthesis) and a vacuum halo profile that is physically motivated.
2. It requires no new particles. After 40 years of null results from direct detection [22, 23], this is significant.
3. It provides physical meaning for the ISO halo parameters: ρ_0 reflects baryonic mass; r_c reflects disk scale length.

4. It naturally resolves the cusp–core problem through $K_{r_c}(T)$: dwarf galaxies have extended vacuum cores, massive galaxies have compact cores.
5. It predicts that $M_{\text{dark}}/M_{\text{vis}}$ evolves with redshift, testable with ALMA [95–97] and JWST [98] at $z > 2$.

11.3 Testable predictions

- (i) The vacuum profile predicts a sharper truncation beyond r_c than NFW. Deep HI observations extending to $R > 10 R_d$ could distinguish the two profiles [99].
- (ii) K_{r_c} predicts specific core sizes for dwarfs ($r_c/R_d \approx 2$) testable with THINGS [91, 93] and LITTLE THINGS [92, 94] data.
- (iii) At $z > 3$, rotation curves should show comparable flatness but different decomposition (more CGM, less dead stars), testable with ALMA kinematics [95, 97].

12 Conclusions

We have applied the α LGQV framework to rotation curves of five SPARC galaxies spanning the full Hubble sequence. The isothermal vacuum profile provides fits of quality comparable to or better than NFW at equal parameter count (2 free parameters each), and dramatically outperforms MOND. Key results:

For NGC 3198: $\chi^2_\nu = 1.19$ (vacuum) vs. 1.40 (NFW) vs. 9.92 (MOND with optimized Υ_d). The predicted vacuum core radii $r_c = K_{r_c} R_d$ agree with fitted values within 10–14% for two galaxies but show 30–63% discrepancies for the other three, indicating that the morphological coefficient prescription requires further refinement.

The framework’s physical motivation—that known baryonic matter plus vacuum effects might account for “missing mass”—is worth exploring. However, the current version has significant limitations: the vacuum normalization is not yet fully predictive, the screening mechanism is not independently verified, and the framework-level parameters ($K(T)$, f_b , f_{dead}) represent semi-constrained choices rather than first-principles derivations.

Addressing these limitations—particularly completing the screening derivation and testing the model on the full SPARC sample of 175 galaxies with automated fitting—is the subject of ongoing work.

Data Availability

The SPARC database is publicly available [38]. An interactive calculator implementing Eq. (8) is at https://boriskruger.github.io/publicationsiir/alphalgqv_calculator.html. The full fitting code is provided in Appendix A.

Acknowledgments

This work uses the SPARC database [38].

A Computational Methods

The rotation curve fitting was performed using Python 3.11 with `scipy.optimize.minimize` (Nelder–Mead) for nonlinear χ^2 minimization. The baryonic velocity components V_{disk} , V_{bul} , V_{gas} were taken directly from the SPARC database files at $\Upsilon = 1$, then rescaled by $\Upsilon_d = 0.5$ and $\Upsilon_b = 0.7$.

Three models were fitted to each galaxy:

1. **ISO vacuum profile** (2 parameters: ρ_0 , r_c). The vacuum contribution is $v_{\text{vac}}^2(R) = 4\pi G \rho_0 r_c^2 [1 - (r_c/R) \arctan(R/r_c)]$. The total model velocity is $v_{\text{mod}} = \sqrt{(1 + 2\alpha) v_{\text{bar}}^2 + v_{\text{vac}}^2}$ with $\alpha = 0.005$ (screened).
2. **NFW profile** (2 parameters: V_{200} , c). $v_{\text{NFW}}^2(R) = V_{200}^2 [\ln(1+x) - x/(1+x)] / (x g_c)$ where $x = R/r_s$, $r_s = r_{200}/c$, $r_{200} = V_{200}/(10H_0)$, $g_c = \ln(1+c) - c/(1+c)$.
3. **MOND** (0 free parameters). Using the simple interpolation function $\nu(y) = 1/(1 - e^{-\sqrt{y}})$ with $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$: $g_{\text{MOND}} = g_{\text{bar}} \cdot \nu(g_{\text{bar}}/a_0)$.

For the ISO and NFW fits, we used a grid of initial conditions (12×12 for ISO in log-space, 6×7 for NFW) to avoid local minima, selecting the globally best χ^2 .

The full Python code is provided below.

```
#!/usr/bin/env python3
"""Rotation curve fitting: ISO vacuum, NFW, MOND.
LGQV framework applied to SPARC galaxies."""

import numpy as np
from scipy.optimize import minimize

# Physical constants
YD = 0.5 # disk mass-to-light ratio (3.6 um)
YB = 0.7 # bulge mass-to-light ratio
G = 4.302e-6 # G in kpc (km/s)^2 / Msun
alpha = 0.005 # screened coupling constant

def v_bar2(Vgas, Vdisk, Vbul, Yd=0.5, Yb=0.7):
    """Baryonic velocity squared from SPARC.

    SPARC provides Vdisk, Vbul at Y=1.
    Vgas includes 1.33x for helium.
    Sign of Vgas preserved (beam smearing).
    Yd, Yb can be varied for MOND optimization.
    """
    return (Yd * Vdisk**2
            + Yb * Vbul**2
            + np.abs(Vgas) * Vgas)
```

```

# ===== ISO VACUUM PROFILE =====
def v_iso(R, rho0, rc):
    """Pseudo-isothermal vacuum halo velocity.

    rho_vac(r) = rho0 / (1 + (r/rc)^2)
    v^2(R) = 4*pi*G*rho0*rc^2
              * [1 - (rc/R)*arctan(R/rc)]
    """
    return np.sqrt(
        4*np.pi*G*rho0*rc**2
        * (1 - (rc/R)*np.arctan(R/rc))
    )

def fit_iso(R, Vobs, eV, Vgas, Vdisk, Vbul):
    """Fit ISO profile. Returns rho0, rc, chi2_nu.

    Fits in log-space with 12x12 initial grid.
    """
    vb2 = v_bar2(Vgas, Vdisk, Vbul)
    vb = np.sqrt(np.maximum(vb2*(1+2*alpha), 0))

    def chi2(p):
        log_rho0, log_rc = p
        rho0 = 10**log_rho0
        rc = 10**log_rc
        if rc < 0.05 or rc > 500:
            return 1e15
        vh = v_iso(R, rho0, rc)
        vm = np.sqrt(vb**2 + vh**2)
        return np.sum(((Vobs - vm)/eV)**2)

    best = 1e15; bp = None
    for lr in np.linspace(-4, 0, 12):
        for lrc in np.linspace(-0.5, 2.5, 12):
            try:
                r = minimize(chi2, [lr, lrc],
                             method='Nelder-Mead',
                             options={'maxiter': 10000,
                                       'xatol': 1e-8,
                                       'fatol': 1e-8})
                if r.fun < best:
                    best = r.fun
                    bp = r.x
            except:
                pass

```

```

    if bp is None:
        return 0.01, 1.0, 999
    rho0 = 10**bp[0]
    rc = 10**bp[1]
    Ndof = max(len(R) - 2, 1)
    return rho0, rc, best/Ndof

# ===== NFW PROFILE =====
def fit_nfw(R, Vobs, eV, Vgas, Vdisk, Vbul):
    """Fit NFW profile. Returns chi2_nu.

    Parameters: V200 (km/s), c (concentration).
    Same (1+2*alpha) baryonic enhancement as ISO
    for strict fairness between models.
    """
    vb2 = v_bar2(Vgas, Vdisk, Vbul)
    vb = np.sqrt(np.maximum(vb2*(1+2*alpha), 0))

    def v_nfw(R, V200, c):
        r200 = V200 / (10 * 0.072) # H0=72
        rs = r200/c; x = R/rs
        gc = np.log(1+c) - c/(1+c)
        return V200 * np.sqrt(
            (np.log(1+x) - x/(1+x)) / (x*gc))

    def chi2(p):
        V200, c = p
        if V200<10 or V200>500 or c<1 or c>50:
            return 1e15
        vh = v_nfw(R, V200, c)
        vm = np.sqrt(vb**2 + vh**2)
        return np.sum(((Vobs - vm)/eV)**2)

    best = 1e15; bp = None
    for v2 in [50,80,100,130,160,200]:
        for cc in [3,5,8,12,15,20,30]:
            try:
                r = minimize(chi2, [v2, cc],
                    method='Nelder-Mead',
                    options={'maxiter': 10000})
                if r.fun < best:
                    best = r.fun
                    bp = r.x
            except:

```

```

        pass

    if bp is None:
        return 999
    return best/max(len(R)-2, 1)

# ===== MOND =====
def fit_mond(R, Vobs, eV, Vgas, Vdisk, Vbul,
            a0=1.2e-10):
    """MOND with optimized Y_d (1 free parameter).

    Simple interpolation function:
    nu(y) = 1 / (1 - exp(-sqrt(y)))
    Y_d is optimized for fair comparison.
    """
    def chi2_mond(Yd_opt):
        Yd_opt = Yd_opt[0]
        if Yd_opt < 0.1 or Yd_opt > 1.5:
            return 1e15
        vb2 = v_bar2(Vgas, Vdisk, Vbul,
                     Yd=Yd_opt, Yb=0.7)
        vb2p = np.maximum(vb2, 1e-6)
        R_m = R * 3.086e19 # kpc -> m
        g_bar = vb2p * 1e6 / R_m
        g_bar = np.maximum(g_bar, 1e-15)
        g_mond = g_bar / (1 - np.exp(
            -np.sqrt(g_bar/a0)))
        vm2 = g_mond * R_m / 1e6
        vm = np.sqrt(np.maximum(vm2, 0))
        return np.sum(((Vobs-vm)/eV)**2)

    best = 1e15; bp = None
    for yd in [0.2,0.3,0.4,0.5,0.6,0.7,0.8,1.0]:
        try:
            r = minimize(chi2_mond, [yd],
                        method='Nelder-Mead',
                        options={'maxiter': 5000})
            if r.fun < best:
                best = r.fun
                bp = r.x[0]
        except: pass
    Ndof = max(len(R)-1, 1) # 1 free param
    return best/Ndof, bp

```

B NGC 3198 SPARC Data

Table 5 provides the full SPARC data for NGC 3198 used in this paper. The velocity components V_{disk} and V_{bul} are given at $\Upsilon = 1 M_{\odot}/L_{\odot}$; they are rescaled by $\sqrt{\Upsilon_d}$ and $\sqrt{\Upsilon_b}$ in the fitting code. V_{gas} includes the $1.33\times$ helium correction. Data for the remaining four galaxies are available in the SPARC database [38].

Table 5: NGC 3198: representative rows from the SPARC rotation curve data (43 points total). R in kpc, all velocities in km s^{-1} . The full dataset is available in the SPARC database [38].

R	v_{obs}	σ_v	V_{gas}	V_{disk}	V_{bul}
0.32	24.4	35.9	0.00	63.28	0
0.64	43.3	16.3	0.00	73.66	0
0.96	45.5	16.1	0.00	78.98	0
1.28	58.5	15.4	0.35	82.70	0
1.61	68.8	7.61	0.15	84.22	0
3.21	100.0	5.31	−1.14	93.81	0
5.46	134.0	2.36	7.16	133.15	0
5.78	137.0	0.89	8.31	136.45	0
7.06	146.0	1.57	12.87	140.94	0
10.04	152.0	2.00	23.68	128.10	0
14.05	157.0	2.00	34.48	118.12	0
20.05	154.0	2.00	39.83	96.40	0
30.08	146.0	2.00	45.29	76.07	0
44.08	149.0	3.00	47.84	61.63	0

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